

Robust Speech Recognition via Large-Scale Weak Supervision

PROBLEMS THEY TRYING TO SOLVE

Old speech recognition systems had problems:

- Needed fine-tuning for each dataset
- Worked badly in noisy environments
- Poor robustness
- Weak out-of-distribution performance
- Mostly English-only
- Could fail with accents and real-world audio

Example:

A model trained on clean speech may fail in:

- meetings
- phone calls
- crowd noise
- YouTube videos

The paper says current systems overfit datasets.

MAIN IDEA

Their idea: **Scale weakly supervised training**

Instead of:

- small clean human-labeled datasets

Use:

- MASSIVE internet audio + transcripts

Even if transcripts are noisy.

This is called: **Weak Supervision**

WHAT IS WHISPER?

Whisper is:

- an encoder-decoder Transformer
- sequence-to-sequence model
- multitask speech model

- multilingual speech model

It can do:

- ✓ Speech recognition
- ✓ Translation
- ✓ Language identification
- ✓ Voice activity detection
- ✓ Timestamp prediction

DATASET

They collected: **680,000 hours of labeled audio**

Breakdown:

- 438k hours English
- 117k hours non-English
- 125k hours translation data

Languages:

- 96+ languages

This is one of the biggest speech datasets ever.

HOW THEY CLEANED DATA

Internet subtitles are noisy. So they filtered data.

They removed:

- machine-generated captions
- bad transcripts
- mismatched language audio
- duplicated transcripts

Methods used:

a) Heuristics

Detect:

- ALL CAPS text
- lowercase-only captions

- missing punctuation

b) Language detection

Check whether spoken language matches transcript language.

c) Fuzzy deduplication

Remove duplicates.

AUDIO PREPROCESSING

Audio processing pipeline:

1. Resample to 16kHz
2. Convert into: **Log-Mel Spectrogram**
3. Use 30-second audio chunks

The model does NOT directly use raw waveform.

MODEL ARCHITECTURE

They use: **Encoder-Decoder Transformer**

Encoder

Input: Log-Mel spectrogram

Encoder extracts: audio features

Uses:

- convolution layers
- Transformer blocks

Decoder

Decoder predicts:

- text tokens

It is autoregressive -predicts next token one-by-one

IMPORTANT MODEL SIZES

Model	Parameters
Tiny	39M
Base	74M
Small	244M
Medium	769M
Large	1550M

TOKENIZER

They used: **Byte Pair Encoding (BPE)**

Same idea as GPT-2 tokenizer.

Used for:

- multilingual text processing

MULTITASK FORMAT

This is a key contribution. They use special tokens.

Example tokens:

- <|startoftranscript|>
- <|transcribe|>
- <|translate|>
- <|nospeech|>

These tokens tell the model:

- what task to do
- what language
- whether timestamps are needed

This allows ONE MODEL to perform many tasks.

WHAT TASKS CAN WHISPER DO?

a) Speech Recognition

Audio → same language text

Ex: Sinhala speech → Sinhala text

b) Translation

Audio → English text

Ex: Spanish speech → English text

c) Language Identification Identify spoken language.

d) Voice Activity Detection Detect whether speech exists.

e) Timestamp Prediction Predict speech timing.

TRAINING DETAILS

Training used:

- AdamW optimizer
- FP16 training
- data parallelism
- gradient clipping

Batch size:

- 256 segments

Training duration:

- about 2–3 passes over dataset

No heavy augmentation used because dataset diversity itself was huge.

WHAT IS ZERO-SHOT?

Zero-shot learning

Means: Model works on new datasets WITHOUT fine-tuning.

Ex: Train on internet audio → directly test on unseen dataset.

Whisper performs very well in zero-shot settings. This is one of the paper's biggest achievements.

WHY WHISPER IS IMPORTANT

Traditional ASR:

- dataset-specific
- fragile

Whisper:

- generalized
- robust
- multilingual
- handles noise better

PERFORMANCE RESULTS

Whisper was compared against:

- wav2vec 2.0
- XLS-R
- mSLAM
- commercial ASR systems
- humans

It performed:

- near human-level
- strong robustness
- better generalization

ROBUSTNESS TO NOISE

They tested:

- white noise
- pub noise

Whisper handled noisy audio much better than many previous models.

Especially:

- low SNR conditions

LONG-FORM TRANSCRIPTION

Problem:

Whisper only sees:

- 30-second chunks

Solution:

- buffered transcription
- timestamp shifting
- beam search
- temperature fallback

This helped:

- long podcasts
- meetings
- TED talks

MULTILINGUAL LEARNING

They found:

More training data for a language → better performance.

Low-resource languages still perform worse.

MULTITASK LEARNING RESULT

Initially: multitask learning hurts small models

But: large models benefit from multitask learning

This is called: **Positive Transfer**

LIMITATIONS

a) Hallucinations Model may generate unrelated text.

b) Repetition loops Can repeat sentences.

c) Weak low-resource language performance

d) Long-form decoding issues

e) Translation data noise

FUTURE WORK

Future improvements:

- better decoding
- more low-resource language data
- fine-tuning research
- stronger language models
- auxiliary objectives

MOST IMPORTANT TERMS YOU MUST KNOW

Term	Meaning
ASR	Automatic Speech Recognition
Weak Supervision	Noisy labels from internet
Zero-shot	No fine-tuning
Transformer	Deep learning architecture
Encoder	Extracts audio features
Decoder	Generates text
Multitask Learning	One model for many tasks
Multilingual	Multiple languages
WER	Word Error Rate
BLEU	Translation accuracy metric
Log-Mel Spectrogram	Audio representation
BPE	Byte Pair Encoding

